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$$V^+(x) = \frac{1}{4}\omega_x^2 x^2, \quad -\infty < x < \infty, \quad (1)$$

$$\psi_n^+(x) \propto e^{-\frac{\omega_x}{4}x^2} H_n \left(\sqrt{\frac{\omega_x}{2}} x \right)$$

$$E_n^+ = \left(n + \frac{1}{2} \right) \omega_x, \quad n = 0, 1, 2, \dots \quad (2)$$

$$V^-(x, m) = V^+(x) - 2 \left[\frac{\mathcal{H}_m'' \left(\sqrt{\frac{\omega_x}{2}} x \right)}{\mathcal{H}_m \left(\sqrt{\frac{\omega_x}{2}} x \right)} - \left[\frac{\mathcal{H}_m' \left(\sqrt{\frac{\omega_x}{2}} x \right)}{\mathcal{H}_m \left(\sqrt{\frac{\omega_x}{2}} x \right)} \right]^2 + \frac{\omega_x}{2} \right], \quad -\infty < x < \infty \quad (3)$$

$$\psi_0^-(x, m) = \phi_m^{-1}(x) \propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m \left(\sqrt{\frac{\omega_x}{2}} x \right)}, \quad m = 0, 2, 4, \dots \quad (4)$$

$$\text{and } \psi_{n+1}^-(x, m) \propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m \left(\sqrt{\frac{\omega_x}{2}} x \right)} \hat{H}_{n+1, m} \left(\sqrt{\frac{\omega_x}{2}} x \right), \quad n = 0, 1, 2, \dots$$

$$\hat{H}_{n+1, m}(z) = \left[\mathcal{H}_m(z) H_{n+1}(z) + H_n(z) \frac{d}{dz} \mathcal{H}_m(z) \right], \quad (5)$$

$$E_{n+1, m}^- = E_{n, m}^+ = E_n^+ - \epsilon_m = (n + m + 1) \omega_x \quad \text{with} \quad E_{0, m}^- = 0. \quad (6)$$

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$$V_h^+(x) = \begin{cases} \frac{1}{4}\omega_x^2 x^2 & , \quad x > 0 \\ \infty & , \quad x \leq 0. \end{cases} \quad (7)$$

$$\psi_{h, n}^+(x, \alpha) \propto x^{\alpha + \frac{1}{2}} e^{-\frac{\omega_x}{4}x^2} L_n^{(\alpha)} \left(\frac{\omega_x}{2} x^2 \right); \quad n = 0, 1, 2, \dots \quad (8)$$

$$\phi_{h, m}(x, \alpha) \propto x^{\alpha + \frac{1}{2}} e^{\frac{\omega_x}{4}x^2} L_m^{(\alpha)} \left(-\frac{\omega_x}{2} x^2 \right), \quad m = 0, 1, 2, \dots \quad (9)$$

$$E_{h, n}^+(\alpha) = (2n + 1 + \alpha) \omega_x \quad \text{and} \quad \epsilon_{h, m}(\alpha) = -(2m + 1 + \alpha) \omega_x \quad (10)$$

$$E_{h, n, m}^+(\alpha) = E_{h, n}^+(\alpha) - \epsilon_{h, m}(\alpha) = 2(n + m + \alpha + 1) \omega_x. \quad (11)$$

$$V_h^-(x, m, \frac{1}{2}) = V_h^+(x) - 2 \left(\frac{\left[x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right) \right]''}{x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} - \left(\frac{\left[x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right) \right]'}{x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} \right)^2 + \frac{\omega_x}{2} \right); \quad m = 0, 1, 2, \dots \quad (12)$$

$$V_h^-(x, m, -\frac{1}{2}) = V_h^+(x) - 2 \left(\frac{L_m^{(-\frac{1}{2})''}(-\frac{\omega_x}{2}x^2)}{L_m^{(-\frac{1}{2})}(-\frac{\omega_x}{2}x^2)} - \left(\frac{L_m^{(-\frac{1}{2})'}(-\frac{\omega_x}{2}x^2)}{L_m^{(-\frac{1}{2})}(-\frac{\omega_x}{2}x^2)} \right)^2 + \frac{\omega_x}{2} \right); \quad m = 0, 1, 2, \dots \quad (13)$$

$$\psi_{h,n}^-(x, m, \alpha) \propto x^{\alpha+\frac{3}{2}} e^{-\frac{\omega_x}{4}x^2} \frac{\hat{L}_{n,m}^{(\alpha)}(\frac{\omega_x}{2}x^2)}{L_m^{(\alpha)}(-\frac{\omega_x}{2}x^2)} \quad (14)$$

$$\hat{L}_{n,m}^{(\alpha)}(s) = L_m^{(\alpha)}(-s)L_n^{(\alpha+1)}(s) + L_{m-1}^{(\alpha+1)}(-s)L_n^{(\alpha)}(s). \quad (15)$$

$$E_{h,n,m}^-(\alpha) = E_{h,n,m}^+(\alpha) = 2(n+m+1+\alpha)\omega_x. \quad (16)$$

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2.1

$$V^+(x, y) = \frac{1}{4}\omega_x^2x^2 + \frac{1}{4}\omega_y^2y^2; \quad -\infty < x < \infty, -\infty < y < \infty \quad (17)$$

$$\begin{aligned} \psi_{n_1,n_2}^+(x, y) &\propto \psi_{n_1}^+(x)\psi_{n_2}^+(y) \\ &\propto e^{-\frac{\omega_x x^2 + \omega_y y^2}{4}} H_{n_1} \left(\sqrt{\frac{\omega_x}{2}} x \right) H_{n_2} \left(\sqrt{\frac{\omega_y}{2}} y \right) \quad n_1, n_2 = 0, 1, 2, \dots, \end{aligned} \quad (18)$$

$$E_{n_1,n_2}^+ = E_{n_1}^+ + E_{n_2}^+ \quad (19)$$

$$\phi_{m_1,m_2}(x, y) = \prod_{k=1}^2 \phi_{m_k}(x_k) \quad \text{where} \quad (x_1, x_2) \rightarrow (x, y). \quad (20)$$

$$\phi_{m_1,m_2}(x, y) \propto e^{\frac{\omega_x x^2 + \omega_y y^2}{4}} H_{m_1} \left(i\sqrt{\frac{\omega_x}{2}} x \right) H_{m_2} \left(i\sqrt{\frac{\omega_y}{2}} y \right) \quad (21)$$

$$V^-(x, y, m_1, m_2) = V^-(x, m_1) + V^-(y, m_2) \quad (22)$$

$$\begin{aligned} \psi_{0,0}^-(x, y, m_1, m_2) &= \phi_{m_1}^{-1}(x)\phi_{m_2}^{-1}(y), \\ \text{and } \psi_{n_1+1,n_2+1}^-(x, y, m_1, m_2) &= \psi_{n_1+1}^-(x, m_1)\psi_{n_2+1}^-(y, m_2), \quad n_1, n_2 = 0, 1, 2, \dots \end{aligned} \quad (23)$$

$$E_{n_1+1,n_2+1,m_1,m_2}^- = [(n_1+m_1+1)\omega_x + (n_2+m_2+1)\omega_y] \quad \text{and} \quad E_{0,0,m_1,m_2}^- = 0. \quad (24)$$

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$$\psi_{h,n_1,n_2}^+(x,y,\alpha,\beta) \propto x^{\alpha+\frac{1}{2}}y^{\beta+\frac{1}{2}}e^{-\frac{\omega_x x^2+\omega_y y^2}{4}}L_{n_1}^{(\alpha)}\left(\frac{\omega_x}{2}x^2\right)L_{n_2}^{(\beta)}\left(\frac{\omega_y}{2}y^2\right); \quad n_1, n_2 = 0, 1, 2, \dots \quad (25)$$

$$\phi_{h,m_1,m_2}(x,y,\alpha,\beta) \propto x^{\alpha+\frac{1}{2}}y^{\beta+\frac{1}{2}}e^{\left(\frac{\omega_x x^2+\omega_y y^2}{4}\right)}L_{m_1}^{(\alpha)}\left(-\frac{\omega_x}{2}x^2\right)L_{m_2}^{(\beta)}\left(-\frac{\omega_y}{2}y^2\right); \quad m_1, m_2 = 0, 1, 2, \dots \quad (26)$$

$$E_{h,n_1,n_2,\alpha,\beta}^+ = [(2n_1 + \alpha + 1)\omega_x + (2n_2 + \beta + 1)\omega_y] - \epsilon_{h,m_1,m_2}, \quad (27)$$

$$V_h^-(x,y,m_1,m_2,\frac{1}{2}) = V_h^-(x,m_1,\frac{1}{2}) + V_h^-(y,m_2,\frac{1}{2}) \quad (28)$$

$$\psi_{h,n_1,n_2}^-(x,y,m_1,m_2,\frac{1}{2}) \propto \psi_{n_1}^-(x,m_1,\frac{1}{2})\psi_{n_2}^-(y,m_2,\frac{1}{2}) \quad (29)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1,m_2,\frac{1}{2}) = [(2n_1 + 2m_1 + 3)\omega_x + (2n_2 + 2m_2 + 3)\omega_y] \quad (30)$$

$$V_h^-(x,y,m_1,m_2,-\frac{1}{2}) = V_h^-(x,m_1,-\frac{1}{2}) + V_h^-(y,m_2,-\frac{1}{2}) \quad (31)$$

$$\psi_{h,n_1,n_2}^-(x,y,m_1,m_2,-\frac{1}{2}) \propto \psi_{h,n_1}^-(x,m_1,-\frac{1}{2})\psi_{h,n_2}^-(y,m_2,-\frac{1}{2}) \quad (32)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1,m_2,\frac{1}{2}) = [(2n_1 + 2m_1 + 1)\omega_x + (2n_2 + 2m_2 + 1)\omega_y] . \quad (33)$$

$$V_h^-(x,y,m_1,m_2,\frac{1}{2},-\frac{1}{2}) = V_h^-(x,m_1,\frac{1}{2}) + V_h^-(y,m_2,-\frac{1}{2}) \quad (34)$$

$$\psi_{h,n_1,n_2}^-(x,y,m_1,m_2,\frac{1}{2},-\frac{1}{2}) \propto \psi_{h,n_1}^-(x,m_1,\frac{1}{2})\psi_{h,n_2}^-(y,m_2,-\frac{1}{2}) \quad (35)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1,m_2,\frac{1}{2},-\frac{1}{2}) = [(2n_1 + 2m_1 + 3)\omega_x + (2n_2 + 2m_2 + 1)\omega_y] \quad (36)$$

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$$V_{fh}^-(x,y,m_1,m_2,\beta) = V^-(x,m_1) + V_h^-(y,m_2,\beta), \quad (37)$$

$$\psi_{fh,0,0}^-(x,y,m_1,m_2,\beta) \propto \psi_0^-(x,m_1)\psi_{h,0}^-(y,m_2,\beta) \quad (38)$$

$$\psi_{fh,n_1+1,n_2}^-(x,y,m_1,m_2,\beta) \propto \psi_{n_1+1}^-(x,m_1)\psi_{h,n_2}^-(y,m_2,\beta); \quad (39)$$

$$E_{fh,0,0}^-(m_2,\beta) = 2(m_2 + \beta + 1)\omega_y. \quad (40)$$

$$\begin{aligned} E_{fh,n_1+1,n_2}^-(m_1,m_2,\beta) &= E_{n_1}^+(m_1) + E_{h,n_2}^+(m_2,\beta) \\ &= (n_1 + m_1 + 1)\omega_x + 2(n_2 + m_2 + \beta + 1)\omega_y \end{aligned} \quad (41)$$

$$V_{fh}^-(x, y, m_1, m_2, 1/2) = V^-(x, m_1) + V_h^-(y, m_2, \frac{1}{2}) \quad (42)$$

$$\psi_{fh,0,0}^-(m_1, m_2, \frac{1}{2}) \propto \psi_0^-(x, m_1) \psi_{h,0}^-(y, m_2, \frac{1}{2}) \quad (43)$$

$$\psi_{n_1+1,n_2}^-(m_1, m_2, \frac{1}{2}) \propto \psi_{n_1+1}^-(x, m_1) \psi_{h,n_2}^-(y, m_2, \frac{1}{2}) \quad (44)$$

$$E_{0,0}^-(m_1, m_2, \frac{1}{2}) = (2m_2 + 3) \omega_y \quad (45)$$

$$E_{n_1+1,n_2}^-(m_1, m_2, \frac{1}{2}) = (n_1 + 2m_1 + 1) \omega_x + (2n_2 + 2m_2 + 3) \omega_y \quad (46)$$

$$V_{fh}^-(x, y, m_1, m_2, -1/2) = V^-(x, m_1) + V_h^-(y, m_2, -\frac{1}{2}) \quad (47)$$

$$\psi_{fh,0,0}^-(m_1, m_2, -\frac{1}{2}) \propto \psi_0^-(x, m_1) \psi_{h,0}^-(y, m_2, -\frac{1}{2}) \quad (48)$$

$$\psi_{n_1+1,n_2}^-(m_1, m_2, -\frac{1}{2}) \propto \psi_{n_1+1}^-(x, m_1) \psi_{h,n_2}^-(y, m_2, -\frac{1}{2}) \quad (49)$$

$$E_{0,0}^-(m_1, m_2, -\frac{1}{2}) = (2m_2 + 1) \omega_y \quad (50)$$

$$E_{n_1+1,n_2}^-(m_1, m_2, -\frac{1}{2}) = (n_1 + 2m_1 + 1) \omega_x + (2n_2 + 2m_2 + 1) \omega_y \quad (51)$$

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3.1

$$V^+(x, y, z) = \frac{1}{4} \omega^2 (x^2 + y^2) + \frac{1}{4} \omega_z^2 z^2; \quad -\infty < x, y, z < \infty \quad (52)$$

$$V^+(r, z) = \frac{1}{4} \omega^2 r^2 + \frac{1}{4} \omega_z^2 z^2 \quad (53)$$

$$\left[- \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \right) + V^+(r, z) \right] \psi^+(r, \phi, z) = E \psi^+(r, \phi, z), \quad (54)$$

$$\psi^+(r, \phi, z) = \frac{e^{i\gamma\phi} \zeta(r, z)}{r^{\frac{1}{2}}}. \quad (55)$$

$$\zeta(r, z) = R(r) Z(z). \quad (56)$$

$$\left[-\frac{\partial^2}{\partial r^2} - \frac{\partial^2}{\partial z^2} + V_{eff}^+(r, z) \right] \zeta(r, z) = E \zeta(r, z), \quad (57)$$

$$V_{eff}^+(r, z) = V^+(r, z) + \frac{\gamma^2 - 1/4}{r^2} \quad \text{with} \quad \gamma = 0, \pm 1, \pm 2, \dots \quad (58)$$

$$R(r) \rightarrow R_{n_1}(r) \propto r^{|\gamma| + \frac{1}{2}} e^{-\frac{\omega}{4} r^2} L_{n_1}^{(|\gamma|)} \left(\frac{1}{2} \omega r^2 \right); n_1 = 0, 1, 2, \dots \quad (59)$$

$$Z(z) \rightarrow Z_{n_2}(z) \propto e^{-\frac{\omega_z z^2}{4}} H_{n_2} \left(\sqrt{\frac{\omega_z}{2}} z \right); n_2 = 0, 1, 2, \dots \quad (60)$$

$$E \rightarrow E_{n_1, n_2} = (2n_1 + |\gamma| + 1)\omega + \left(n_2 + \frac{1}{2} \right) \omega_z. \quad (61)$$

$$E_{n_1, n_2}^+ = E_{n_1, n_2} - \epsilon_{m_1, m_2} = 2(n_1 + m_1 + |\gamma| + 1)\omega + (n_2 + m_2 + 1)\omega_z. \quad (62)$$

$$\phi_{m_1}(r) \propto r^{|\gamma| + \frac{1}{2}} e^{\frac{\omega}{4} r^2} L_{m_1}^{(|\gamma|)} \left(-\frac{1}{2} \omega r^2 \right); \quad m_1 = 0, 1, 2, \dots \quad (63)$$

$$\phi_{m_2}(z) \propto e^{\frac{\omega_z}{4} z^2} \mathcal{H}_{m_2} \left(\sqrt{\frac{\omega_z}{2}} z \right); \quad m_2 = 0, 2, 4, \dots \quad \text{Since } z \text{ axis is full line.}$$

$$\phi_{m_1, m_2}(r, z) = \phi_{m_1}(r) \phi_{m_2}(z) \quad (64)$$

$$\begin{aligned} A &= \frac{\partial}{\partial r} + \frac{\partial \log(\phi_{m_1, m_2}(r, z))}{\partial r} + i \frac{\partial}{\partial z} + i \frac{\partial \log \phi_{m_1, m_2}(r, z)}{\partial z} \\ \text{and } A^\dagger &= -\frac{\partial}{\partial r} + \frac{\partial \log(\phi_{m_1, m_2}(r, z))}{\partial r} + i \frac{\partial}{\partial z} - i \frac{\partial \log \phi_{m_1, m_2}(r, z)}{\partial z} \end{aligned} \quad (65)$$

$$\begin{aligned} V_{eff}^-(m_1, m_2, r, z) &= \frac{1}{4} \omega^2 r^2 + \frac{\gamma^2 - \frac{1}{4}}{r^2} + \frac{2|\gamma|}{r^2} - (2m + 1)\omega + \frac{1}{r^2} \\ &\quad + \frac{2r^2 \omega^2 L_{m_1-1}^{(|\gamma|+1)} \left(-\frac{\omega r^2}{2} \right)^2}{L_{m_1}^{(|\gamma|)} \left(-\frac{\omega r^2}{2} \right)^2} \\ &\quad + \frac{\omega \left((2|\gamma| + r^2 \omega) L_{m_1-1}^{(|\gamma|+1)} \left(-\frac{\omega r^2}{2} \right) - r^2 \omega L_{m-2}^{(|\gamma|+2)} \left(-\frac{\omega r^2}{2} \right) \right)}{L_{m_1}^{(|\gamma|)} \left(-\frac{\omega r^2}{2} \right)} \\ &\quad + \frac{1}{2} \omega_z^2 z^2 - 2 \left[\frac{\mathcal{H}_{m_2}'' \left(\sqrt{\frac{\omega_z}{2}} z \right)}{\mathcal{H}_{m_2} \left(\sqrt{\frac{\omega_z}{2}} z \right)} - \left[\frac{\mathcal{H}_{m_2}' \left(\sqrt{\frac{\omega_z}{2}} z \right)}{\mathcal{H}_{m_2} \left(\sqrt{\frac{\omega_z}{2}} z \right)} \right]^2 + \frac{\omega_z}{2} \right] \\ \zeta_{n_1, n_2+1}^-(r, z, m_1, m_2) &\propto r^{|\gamma| + \frac{3}{2}} e^{-\frac{\omega r^2}{4}} \frac{\hat{L}_{n_1, m_1}^{(\gamma)} \left(\frac{1}{2} \omega r^2 \right)}{L_{m_1}^{(\gamma)} \left(-\frac{1}{2} \omega r^2 \right)} e^{-\frac{\omega_z z^2}{2}} \frac{\hat{H}_{n_2+1, m_2} \left(\sqrt{\frac{\omega_z}{2}} z \right)}{\mathcal{H}_{m_2} \left(\sqrt{\frac{\omega_z}{2}} z \right)} \\ E_{n_1, n_2+1}^-(m_1, m_2) &= 2(n_1 + m_1 + |\gamma| + 1)\omega + (n_2 + m_2 + 1)\omega_z \end{aligned} \quad (66)$$

$$\zeta_{0,0}^-(r, z, m_1, m_2) \propto r^{|\gamma|+\frac{3}{2}} \frac{e^{-\frac{\omega r^2}{4}} L_{m_1}^{(|\gamma|+1)}\left(-\frac{1}{2}\omega r^2\right)}{L_{m_1}^{(|\gamma|)}\left(-\frac{1}{2}\omega r^2\right)} \times \frac{e^{-\frac{\omega_z z^2}{4}}}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}$$

and $E_{0,0,m_1,m_2}^- = 2(m_1 + |\gamma| + 1)\omega$ (67)

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$$[-\nabla^2 + V(x)] \psi_n(x) = E \psi_n(x) \quad (68)$$

$$H \psi_n(x) = E \psi_n(x), \quad (69)$$

$$A = \frac{d}{dx} + W(x)$$

and $A^\dagger = -\frac{d}{dx} + W(x)$ (70)

$$H - \epsilon = \begin{cases} A^\dagger A = -\frac{d^2}{dx^2} + V^-(x) - \epsilon, & \text{say } H^- \\ AA^\dagger = -\frac{d^2}{dx^2} + V^+(x) - \epsilon, & \text{say } H^+ \end{cases} \quad (71)$$

$$V^\mp(x) = W(x)^2 \mp W(x)' + \epsilon \quad (72)$$

$$\left[-\frac{d^2}{dx^2} + V^+(x)\right] \psi_0^+(x) = E_0^+ \psi_0^+(x) \quad (73)$$

$$\left[-\frac{d^2}{dx^2} + V^+(x)\right] \phi_m(x) = \epsilon_m \phi_m(x) \quad (74)$$

$$W(x) = \begin{cases} -\frac{d}{dx} \ln[\phi_m(x)], & \text{for } \epsilon_m = E_0 \\ +\frac{d}{dx} \ln[\phi_m(x)], & \text{for } \epsilon_m < E_0 \end{cases} \quad (75)$$

$$\psi_0^-(x, m) \propto \phi_m(x) \quad (76)$$

$$\psi_0^-(x, m) \propto \phi_m^{-1}(x) \quad (77)$$

$$(A A^\dagger) A \psi^- = H^+ A \psi^- = E^+ \frac{1}{C_-} \psi^+ \quad (78)$$

$$(A^\dagger A) A^\dagger \psi^+ = H^- A^\dagger \psi^+ = E^- \frac{1}{C_+} \psi^- \quad (79)$$

$$E_{n,m}^+ = E_n^+ - \epsilon_m \quad (80)$$

$$\begin{cases} \psi_{n+1}^-(x, m) = \frac{1}{\sqrt{E_{n,m}^+}} A^\dagger \psi_n^+ & \text{and} & E_{n+1,m}^- = E_{n,m}^+; \quad E_{0,m}^- = 0 & \text{for normalizable } \phi_m^{-1}(x) \\ \psi_n^-(x, m) = \frac{1}{\sqrt{E_{n,m}^+}} A^\dagger \psi_n^+ & \text{and} & E_{n,m}^- = E_{n,m}^+ & \text{for non-normalizable } \phi_m(x) \text{ or } \phi_m^{-1}(x) \end{cases} \quad (81)$$

$$A = \hat{e}^+ \cdot (\vec{\nabla} + \vec{W}), \quad \hat{e}^+ = \sum_{i=1}^D c_i \hat{e}_i \quad (82)$$

$$A^\dagger = (\vec{\nabla}^\dagger + \vec{W}^\dagger) \cdot \hat{e}^{+\dagger}, \quad \hat{e}^{+\dagger} = \sum_{i=1}^D c_i^\dagger \hat{e}_i, \quad (83)$$

$$i_j i_k = -\delta_{jk} + \sum_{l=1}^3 \epsilon_{jkl} i_l, \quad (84)$$

$$V^\mp = \sum_k^D W_k^2 \mp \sum_k^D \frac{\partial W_k}{\partial x_k} + Q_D, \quad (85)$$

$$W_k = - \left| \frac{\partial}{\partial x_k} \ln[\phi_{m_k}(x_k)] \right| \quad (86)$$

$$Q_1 = 0$$

$$Q_2 = i \left(W_x \frac{\partial}{\partial y} - W_y \frac{\partial}{\partial x} \right) \quad (87)$$

$$\text{and } Q_3 = \mp 2 \left[i_1 \left(W_y \frac{\partial}{\partial z} - W_z \frac{\partial}{\partial y} \right) + i_2 \left(W_z \frac{\partial}{\partial x} - W_x \frac{\partial}{\partial z} \right) + i_3 \left(W_x \frac{\partial}{\partial y} - W_y \frac{\partial}{\partial x} \right) \right]$$

$$V^+(x_1, x_2, \dots) = \sum_k^D V_k^+(x_k) \quad (88)$$

$$\psi_{n_1, n_2, \dots}^+(x, y, \dots) = \prod_k \psi_{n_k}^+(x_k). \quad (89)$$

$$E_{n_1, n_2, \dots}^+ = \sum_k^D E_{n_k}^+ \quad \text{where } n_k \in \mathbb{Z} \quad (90)$$

$$V^-(x_1, x_2, \dots, m_1, m_2, \dots) = V^+(x_1, x_2, \dots) - 2 \sum_k^D \left| \frac{\partial^2 \phi_k(x_k)}{\partial x_k^2} \right| + Q_D \quad (91)$$

$$\psi_{n_1, n_2, \dots}^-(x_1, x_2, \dots, m_1, m_2, \dots) = \prod_k \psi_{n_k}^-(x_k, m_k) \quad (92)$$

$$E^-(m_1, m_2, \dots) = \sum_k^D E^-(m_k) \quad (93)$$

$$\psi_{n_1, n_2}(x, y) \propto e^{-\frac{\omega_x}{4} x^2 - \frac{\omega_y}{4} y^2} H_{n_1} \left(\sqrt{\frac{\omega_x}{2}} x \right) H_{n_2} \left(\sqrt{\frac{\omega_y}{2}} y \right) \quad (94)$$